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Inventor(s)

Lou; Jianquan et al.

SIGNAL VIA OFFSET ARRANGEMENT FOR PRINTED CIRCUIT BOARD

Abstract

An apparatus that includes a layer of a printed circuit board and a first pair of signal vias and a second pair of signal vias each formed through the layer. The first pair of signal vias includes a first signal via and a second signal via that are offset along a first axis and aligned along a second axis, perpendicular to the first axis. The second pair of signal vias includes a third signal via and a fourth signal via that are offset along the first axis and aligned along the second axis. The pairs of signal vias are offset along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from a second distance between the third signal via and the second signal via.

Inventors: Lou; Jianquan (Shanghai, CN), Bhobe; Alpesh Umakant (Sunnyvale, CA)

Applicant: Cisco Technology, Inc. (San Jose, CA)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to printed circuit boards (PCBs).

BACKGROUND

[0002] A printed circuit board (PCB) electrically couples various electronic components with one another. For example, a PCB may include multiple layers, each having different electronic components and traces routed along the layers to electrically couple electronic components of the same layer. Additionally, the PCB may include vias that extend between layers to electrically couple electronic components of different layers to one another. For instance, a signal may propagate from a first electronic component of a first layer, through a first trace routed along the first layer, through a via extending from the first layer to a second layer, through a second trace routed along the second layer, and to an electronic component of the second layer. Unfortunately, the PCB may be subject to crosstalk in which transmitted signals interfere with one another. For example, electric fields emitted by vias during signal transmission may overlap with one another to cause via-to-via crosstalk. Interference between signals may increase signal loss, thereby reducing signal integrity during signal transmission.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a perspective cross-sectional view of a printed circuit board (PCB) that includes multiple layers, according to an example embodiment.

[0004] FIG. 2 is a schematic diagram of signal vias in an offset arrangement at a PCB layer, according to an example embodiment.

[0005] FIG. 3 is a schematic diagram of signal vias in an offset arrangement at a PCB layer, according to an example embodiment.

[0006] FIG. 4 is a graph illustrating measurements of crosstalk between signal vias at different offset arrangements, according to an example embodiment.

[0007] FIG. 5 is a schematic diagram of signal vias and ground vias in an offset arrangement at a PCB layer, according to an example embodiment.

[0008] FIG. 6 is a schematic diagram of pins, signal vias, ground vias in an offset arrangement at a PCB layer, according to an example embodiment.

[0009] FIG. 7 is a flowchart of a method of manufacturing a PCB layer with signal vias and ground vias, according to an example embodiment.

DETAILED DESCRIPTION

Overview

[0010] Techniques are provided herein for reducing crosstalk between signal vias. In some aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; a first pair of signal vias formed through the layer, wherein the first pair of signal vias is configured to transmit respective signals, the first pair of signal vias comprises a first signal via and a second signal via offset from one another along a first axis and aligned with one another along a second axis, perpendicular to the first axis; and a second pair of signal vias formed through the layer, wherein the second pair of signal vias is configured to transmit respective signals, the second pair of signal vias comprises a third signal via and a fourth signal via offset from one another along the first axis and aligned with one another along the second axis, and the first pair of signal vias and the second pair of signal vias are offset from one another along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from a second distance between the third signal via and the second signal via.

[0011] According to other aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; and a plurality of signal vias formed through the layer and

configured to transmit respective signals, wherein the plurality of signal vias comprises: a first signal via; a second signal via immediately adjacent to the first signal via along a first axis; a third signal via immediately adjacent to the first signal via along a second axis, perpendicular to the first axis; and a fourth signal via immediately adjacent to the third signal via along the first axis, wherein the first signal via, the second signal via, the third signal via, and the fourth signal via are arranged in a parallelogram configuration, and a first distance between the first signal via and the third signal via is different from a second distance between the second signal via and the third signal via.

[0012] In still other aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; and a plurality of signal vias formed through the layer and configured to transmit respective signals, wherein the plurality of signal vias comprises: a first signal via; a second signal via offset from the first signal via along a first axis and aligned with the first signal via along a second axis, perpendicular to the first axis; a third signal via offset from the first signal via and the second signal via along the first axis and along the second axis; and a fourth signal via offset from the third signal via along the first axis and aligned with the third signal via along the second axis, wherein a first distance between the first signal via and the third signal via, a second distance between the first signal via and the second signal via, and a third distance between the second signal via and the third signal via are different from one another, and the second distance between the first signal via and the second signal via is substantially equal to a fourth distance between the third signal via and the fourth signal via.

Example Embodiments

[0013] Embodiments of the present disclosure are directed to a printed circuit board (PCB) having signal vias formed at a layer of the PCB and arranged to reduce crosstalk. Each signal via may be configured to transmit a respective signal, such between layers of the PCB. As such, components at different layers of the PCB may communicate with one another using the signal transmitted through the signal vias.

[0014] The layer of the PCB may include a first pair of signal vias that has a first signal via and a second signal via offset from one another along a first axis and aligned with one another along a second axis, perpendicular to the first axis. The layer of the PCB may also include a second pair of signal vias that has a third signal via and a fourth signal via offset from one another along the first axis and aligned with one another along the second axis. The first pair of signal vias and the second pair of signal vias are offset from one another along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from a second distance between the third signal via and the second signal via.

[0015] Such an arrangement between the first pair of signal vias and the second pair of signal vias may reduce crosstalk between the pairs of signal vias. For instance, the provided offset between the pairs of signal vias may reduce an overlap of electric fields emitted by the pairs of signal vias during signal transmission. As a result, integrity of the signals transmitted by the pairs of signal vias may be improved.

[0016] With reference made to FIG. 1, depicted therein is a cross-sectional view of a PCB **100** having multiple layers **102**. Each layer **102** may include different electronic components that are electrically coupled to one another. By way of example, a first layer **102A** (e.g., a top layer) may include multiple pads **104** that are exposed to an exterior environment of the PCB **100**. The pads **104** may enable electrical coupling of the PCB **100** to a separate component, such as an integrated circuit (IC) (e.g., an application-specific IC (ASIC)). For example, an interconnect **106**, such as a solder ball, may be used to electrically couple one of the pads **104** to the separate component. A trace **108** may also be electrically coupled to the pad **104** and routed along the first layer **102A** to electrically couple the pad **104**, and therefore the interconnect **106** and the separate component electrically coupled to the pad **104**, to an electronic component (e.g., another IC, a resistor, a transistor, a capacitor, a switch, an inductor, a transformer, a sensor, a diode, a relay) of the first

layer **102A**.

[0017] Moreover, the PCB **100** may include signal vias **110** that extend through multiple layers **102** of the PCB **100** to electrically couple electronic components of different layers **102** to one another. For instance, a first signal via **110A** may be electrically coupled to the pad **104** that is electrically coupled to the interconnect **106**, and the first signal via **110A** may extend from the first layer **102A** to a second layer **102B** (e.g., an inner layer, a mid layer) of the PCB **100**. A trace (not shown) routed along the second layer **102B** may be electrically coupled to the first signal via **110A** and to an electronic component of the second layer **102B**. As such, the first signal via **110A** may help electrically couple the electronic component of the second layer **102B** to the electronic component of the first layer **102A** and/or to the separate component electrically coupled to the pad **104**.

[0018] By way of example, during operation of the PCB **100**, a signal may be transmitted between the electronic component of the first layer **102A**, the electronic component of the second layer **102B**, and/or the separate component electrically coupled to the PCB **100**. For instance, the separate component may transmit the signal to the pad **104** by way of the interconnect **106**, and the signal may propagate through the trace **108** secured/connected to the pad **104** and routed along the first layer **102A** toward the electronic component of the first layer **102A** and/or through the first signal via **110A** toward the second layer **102B**, through the trace routed along the second layer **102B** and electrically coupled to the first signal via **110A**, and toward the electronic component of the second layer **102B**.

[0019] However, an integrity of the signal propagated through the first signal via **110A** may be reduced as a result of electrical field interferences. For example, a second signal via **110B** may extend from the first layer **102A** and may also be configured to transmit a signal. During signal transmission, the signal vias **110** may emit respective electrical fields toward one another. That is, the first signal via **110A** may emit an electrical field toward the second signal via **110B**, and the second signal via **110B** may emit an electrical field toward the via **110A**. Overlap between the electrical fields may cause crosstalk that reduces an integrity of the signals transmitted through the signal vias **110**. Consequently, operation of the PCB **100** using the signals may be affected.

[0020] For this reason, the signal vias **110** may be arranged in a manner to achieve desirable signal integrity. By way of example, the signal vias **110** are offset at particular distances that reduce an overlap in their emitted electrical fields without substantially increasing a total area occupied by the signal vias **110**. As such, the offset arrangement of the signal vias **110** may increase signal integrity while still enabling the signal vias **110** to efficiently occupy the PCB **100**. For example, the signal vias **110** can be densely positioned (e.g., positioned adjacent to one another) at the first layer **102A** without reducing signal integrity.

[0021] FIG. **2** is a schematic diagram of a layer **150** of a PCB, such as the first layer **102A** of the PCB **100**. The layer **150** may include a first pair of signal vias **152** and a second pair of signal vias **154**. The first pair of signal vias **152** may include a first signal via **156** and a second signal via **158** configured to transmit respective signals, and the second pair of signal vias **154** may include a third signal via **160** and a fourth signal via **162** configured to transmit respective signals. For instance, the first signal via **156** and the second signal via **158** are a differential pair configured to transmit related signals (e.g., a positive signal, a negative signal), and the third signal via **160** and the fourth signal via **162** are another differential pair, separate from the first pair of signal vias **152**, also configured to transmit related signals (e.g., a positive signal, a negative signal). The first signal via **156** and the second signal via **158** are offset from and immediately adjacent to one another along a first axis **164** (e.g., a longitudinal axis) and aligned with one another along a second axis **166** (e.g., a lateral axis), perpendicular to the first axis **164**. Similarly, the third signal via **160** and the fourth signal via **162** are offset from and immediately adjacent to one another along the first axis **164** and aligned with one another along the second axis **166**.

[0022] Additionally, the first pair of signal vias **152** and the second pair of signal vias **154** are immediately adjacent to one another along the second axis **166**. That is, the first signal via **156** and

the third signal via **160** may be immediately adjacent to one another along the second axis **166**, and the second signal via **158** and the fourth signal via **162** may be immediately adjacent to one another along the second axis **166**. However, the pairs of signal vias **152**, **154** may be offset from one another along the first axis **164** and along the second axis **166**. The offset between the pairs of signal vias **152**, **154** may cause the third signal via **160** to be positioned at a first distance **168** away from the first signal via **156** and at a second distance **170** away from the second signal via **158** and cause the fourth signal via **162** to be positioned at a third distance **172** away from the first signal via **156** and at a fourth distance **174** away from the second signal via **158**. By way of example, the first distance **168** may be greater than the second distance **170** and the third distance **172** may be greater than the fourth distance **174**. In any case, each of the distances **168**, **170**, **172**, **174** may extend obliquely relative to the first axis **164** and the second axis **166**.

[0023] Additionally, the first signal via **156** and the second signal via **158** may be positioned at a fifth distance **176** away from one another along the first axis **164**, and the third signal via **160** and the fourth signal via **162** may be positioned at a sixth distance **178** away from one another along the first axis **164**. In certain embodiments, the fifth distance **176** and the sixth distance **178** may be substantially equal to one another (e.g., greater than the second distance **170**), and the first distance **168** and the fourth distance **174** may therefore be substantially equal to one another, to arrange the pairs of signal vias **152**, **154** in a parallelogram configuration in which each of the signal vias **156**, **158**, **160**, **162** are positioned at a respective corner of a cooperatively formed parallelogram shape. However, the first distance **168**, the second distance **170**, and the fifth distance **176** may be substantially different from one another, and the second distance **170**, the fourth distance **174**, and the sixth distance **178** may be substantially different from one another. Such positioning of the signal vias **156**, **158**, **160**, **162** may reduce crosstalk therebetween, such as by reducing overlap between their respective electrical fields.

[0024] To position the pairs of signal vias **152**, **154** in the illustrated arrangement, the third signal via **160** may be positioned away from the first signal via **156** along the first axis **164** by an offset distance **180**. As an example, positioning the third signal via **160** at the offset distance **180** away from the first signal via **156** may reduce noise associated with energy having a particular frequency (e.g., below a threshold value) to limit crosstalk. The offset distance **180** may be offset from a midpoint **182** between the first signal via **156** and the second signal via **158** along the first axis **164**. For instance, the third signal via **160** may be positioned more adjacent to the second signal via **158** than to the first signal via **156** along the first axis **164**. However, the exact offset distance **180** may differ for different implementations to limit a desirable amount of crosstalk. For example, different operating parameters/conditions, such as a signal transmission rate, a signal via size, noise frequency, a PCB layer arrangement, and so forth related to the PCB, may adjust the offset distance **180** providing a sufficiently low amount of crosstalk between the signal vias **156**, **158**, **160**, **162**. In any case, the offset distance **180** is established to provide a low overall amount of noise coupling between the signal vias **156**, **158**, **160**, **162**. In other words, the signal vias **156**, **158**, **160**, **162** are arranged such that the amount of noise produced by one of the pairs of signal vias **152**, **154** (i.e., the aggressor pair) and affecting the other of the pairs of signal vias **152**, **154** (i.e., the victim pair) is low (e.g., below a threshold value).

[0025] As an example, during operation, the first signal via **156** may produce a first amount (e.g., a positive amount) of noise coupling with respect to the third signal via **160** and a second amount (e.g., a positive amount) of noise coupling with respect to the fourth signal via **162**. Additionally, the second signal via **158** may produce a third amount (e.g., a negative amount) of noise coupling with respect to the third signal via **160** and a fourth amount of noise coupling (e.g., a negative amount) with respect to the fourth signal via **162**. The offset distance **180** may be established such that a first sum of the first amount of noise coupling plus the third amount of noise coupling is substantially equal to a second sum of the second amount of noise coupling plus the fourth amount of noise coupling. Such an offset distance **180** may be a value between 0.66 millimeters (mm) and

0.75 mm (e.g., between 0.026 inches and 0.03 inches).

[0026] In certain embodiments, the signal vias **156, 158, 160, 162** may be configured to directly mount to (e.g., using a soldering technique) corresponding pins of a separate component, such as an integrated circuit (IC). To this end, the pins of the separate component are arranged in a corresponding manner as the illustrated arrangement of the signal vias **156, 158, 160, 162** (e.g., the pins are offset along the first axis and along the second axis in the same manner) to enable the pins to align with the signal vias **156, 158, 160, 162**. As a result, the signals may be transmitted between the layer **150** and the separate component by way of the signal vias **156, 158, 160, 162**.

[0027] FIG. **3** is a schematic diagram of the layer **150** having the first pair of signal vias **152**, the second pair of signal vias **154**, and a third pair of signal vias **200**. The third pair of signal vias **200** may be immediately adjacent to the second pair of signal vias **154** along the second axis **166**, and the third pair of signal vias **200** may include a fifth signal via **202** and a sixth signal via **204** that are offset from one another along the first axis **164** and aligned with one another along the second axis **166**. Moreover, the third pair of signal vias **200** may be offset from the second pair of signal vias **154** along the first axis **164** and along the second axis **166**. As an example, the third pair of signal vias **200** may be aligned with the first pair of signal vias **152** along the first axis **164**.

[0028] For instance, the fifth signal via **202** may be positioned at the first distance **168** away from the third signal via **160** and at the third distance **172** away from the fourth signal via **162**, and the sixth signal via **204** may be positioned at the second distance **170** away from the third signal via **160** and at the fourth distance **174** away from the fourth signal via **162**. As such, the first pair of signal vias **152** and the third pair of signal vias **200** may be symmetrical about the second pair of signal vias **154** such that the third signal via **160** may be equidistant to the first signal via **156** and to the fifth signal via **202** and equidistant to the second signal via **158** and to the sixth signal via **204**, and the fourth signal via **162** may be equidistant to the first signal via **156** and to the fifth signal via **202** and equidistant to the second signal via **158** and to the sixth signal via **204**.

However, in alternative embodiments, the first pair of signal vias **152** and the third pair of signal vias **200** may not be symmetrical about the second pair of signal vias **154**. As an example, the third pair of signal vias **200** may be aligned with the first pair of signal vias **152** along the first axis **164** but offset along the second axis **166** such that the third signal via **160** is not equidistant to the first signal via **156** and the fifth signal via **202** (e.g., the third signal via **160** is positioned more adjacent to the first signal via **156** than to the fifth signal via **202**). As another example, the third pair of signal vias **200** may not be aligned with the first pair of signal vias along the first axis **164**. Instead, for instance, the third pair of signal vias **200** may be offset from each of the first pair of signal vias **152** and the second pair of signal vias **154** along the first axis **164** such that the first signal via **156**, the third signal via **160**, and the fifth signal via **202** are collinear along an axis extending obliquely to the first axis **164** and the second axis **166**.

[0029] FIG. **4** illustrates a graph **250** depicting a relationship between the offset distance **180** and a measurement of total crosstalk (e.g., a summation of the amounts of noise coupling) between the two pairs of signal vias **152, 154**, with reference to FIGS. **2** and **3**. That is, the graph **250** indicates an amount of crosstalk between the first pair of signal vias **152** and the second pair of signal vias **154** caused by varying offset distances **180** between the first signal via **156** and the third signal via **160** along the first axis **164**. For instance, a first offset distance **180A** in which the first pair of signal vias **152** and the second pair of signal vias **154** are aligned along the first axis **164** may cause a first crosstalk amount **252** (e.g., a high crosstalk amount, a maximum crosstalk amount).

Increasing the offset distance **180** between the first pair of signal vias **152** and the second pair of signal vias **154** along the first axis **164** from the first offset distance **180A** may reduce the amount of crosstalk from the first crosstalk amount **252**. A second offset distance **180B** in which the third signal via **160** is aligned with the midpoint **182** along the first axis **164** (i.e., the third signal via **160** is positioned midway between the first signal via **156** and the second signal via **158** along the first axis **164**) may cause a second crosstalk amount **254** (e.g., an intermediate crosstalk amount).

Further increasing the offset distance **180** between the first pair of signal vias **152** and the second pair of signal vias **154** along the first axis **164** from the second offset distance **180B** may further reduce the amount of crosstalk from the second crosstalk amount **254**. For example, a third offset distance **180C** in which the third signal via **160** is positioned more adjacent to the second signal via **158** than to the first signal via **156** along the first axis **164** may cause a third crosstalk amount **256** (e.g., a low crosstalk amount, a minimum crosstalk amount). Increasing the offset distance **180** between the first pair of signal vias **152** and the second pair of signal vias **154** along the first axis **164** from the third offset distance **180C** may then increase the amount of crosstalk from the third crosstalk amount **256**. By way of example, a fourth offset distance **180D** in which the third signal via **160** is aligned with the second signal via **158** may cause a fourth crosstalk amount **258**, which may be substantially equal to the first crosstalk amount **252**.

[0030] Thus, as shown in the graph **250**, an arrangement in which the third signal via **160** is not positioned midway between the first signal via **156** and the second signal via **158** (e.g., such that the first distance **168** between the first signal via **156** and the third signal via **160**, the second distance **170** between the second signal via **158** and the third signal via **160**, and the fifth distance **176** between the first signal via **156** and the second signal via **158** are different from one another) may reduce crosstalk to the greatest extent. As an example, the third offset distance **180C** may be a value between 0.66 mm and 0.75 mm (e.g., between 0.026 inches and 0.03 inches), whereas the fifth distance **176** between the first signal via **156** and the second signal via **158** may be a value between 0.8 mm and 1 mm (e.g., between 0.027 inches to 0.039 inches). It should be noted that for different implementations, a different graph may depict the relationship between an offset distance and an amount of crosstalk. For instance, the rate at which the amount of crosstalk increases and decreases may vary for the different graphs. However, for each graph, a particular offset distance (e.g., an offset distance misaligned with a corresponding midway point along the first axis **164**) may cause the lowest amount of crosstalk, and pairs of signal vias may be arranged based on the particular offset distance to limit crosstalk and maintain desirable signal integrity.

[0031] FIG. **5** is a schematic diagram of the layer **150** having the first pair of signal vias **152** and the second pair of signal vias **154**. The layer **150** may also include ground vias positioned around the pairs of signal vias **152**, **154** to further reduce crosstalk. For example, the ground vias may block emission of electrical fields (e.g., toward other signal vias at the layer **150**), thereby isolating the electrical fields emitted by the pairs of signal vias **152**, **154** to maintain desirable signal integrity.

[0032] The ground vias may include a first ground via **300** positioned adjacent to the first signal via **156**, offset from the first signal via **156** along the first axis **164**, and aligned with the first signal via **156** along the second axis **166**, as well as a second ground via **302** positioned adjacent to the second signal via **158**, offset from the second signal via **158** along the first axis **164**, and aligned with the second signal via **158** along the second axis **166**. For instance, the first ground via **300** may block an electrical field emitted from the first signal via **156**, and the second ground via **302** may block an electrical field emitted from the second signal via **158**. The ground vias may also include a third ground via **304** positioned adjacent to the third signal via **160**, offset from the third signal via **160** along the first axis **164**, and aligned with the third signal via **160** along the second axis **166**, as well as a fourth ground via **306** positioned adjacent to the fourth signal via **162**, offset from the fourth signal via **162** along the first axis **164**, and aligned with the fourth signal via **162** along the second axis **166**. As such, the third ground via **304** may block an electrical field emitted from the third signal via **160**, and the fourth ground via **306** may block an electrical field emitted from the fourth signal via **162**.

[0033] The first ground via **300** may be positioned at a seventh distance **308** away from the first signal via **156** along the first axis **164**, the second ground via **302** may be positioned at an eighth distance **310** away from the first signal via **156** along the first axis **164**, the third ground via **304** may be positioned at a ninth distance **312** away from the third signal via **160** along the first axis

164, and the fourth ground via **306** may be positioned at a tenth distance **314** away from the fourth signal via **162** along the first axis **164**. In some embodiments, the seventh distance **308**, the eighth distance **310**, the ninth distance **312**, and the tenth distance **314** may be substantially equal to one another. By way of example, the seventh distance **308**, the eighth distance **310**, the ninth distance **312**, and the tenth distance **314** may be substantially equal to the fifth distance **176** and/or the sixth distance **178**. Therefore, the third ground via **304** may be offset from the first ground via **300** along the first axis **164** and along the second axis **166**, and the fourth ground via **306** may be offset from the second ground via **302** along the first axis **164** and along the second axis **166**. In certain embodiments, such a distance may also be substantially equal to an eleventh distance **316** between the first pair of signal vias **152** and the second pair of signal vias **154** along the second axis **166**. For instance, each of the distances **176**, **178**, **308**, **310**, **312**, **314** may be a value between 0.8 mm and 1 mm (e.g., between 0.027 inches to 0.039 inches).

[0034] It should be noted that no ground vias may be positioned between the pairs of signal vias **152**, **154** along the second axis **166**. That is, the offset arrangement between the pairs of signal vias **152**, **154** may sufficiently block overlap between the respective electrical fields emitted by one another to maintain a desirable signal integrity without having to implement additional ground vias between the pairs of signal vias **152**, **154**. As such, the depicted arrangement may reduce or limit a total quantity of ground vias used at the layer **150**, such as to have a **1** to **1** ratio between ground vias and signal vias such that each ground via may be dedicated to block electrical fields emitted from a corresponding one of the signal vias. For this reason, a cost associated with manufacture of the layer **150** (e.g., of the ground vias) may be reduced, and/or a space of the layer **150** may be more efficiently utilized (e.g., to position operational components, such as additional signal vias, rather than more components, such as ground vias, dedicated to reducing noise).

[0035] FIG. **6** is a schematic diagram of a layer **350** of another PCB. The layer **350** may include a first pair of signal vias **352** and a second pair of signal vias **354**. The first pair of signal vias **352** may include a first signal via **356** and a second signal via **358** offset from one another along the first axis **164** and aligned with one another along the second axis **166**. The second pair of signal vias **354** may include a third signal via **360** and a fourth signal via **362** offset from one another along the first axis **164** and aligned with one another along the second axis **166**. Moreover, the first pair of signal vias **352** and the second pair of signal vias **354** may be offset from one another along both the first axis **164** and the second axis **166**, such as in a similar arrangement as the pairs of signal vias **152**, **154** of the layer **150**, to reduce crosstalk between the pairs of signal vias **352**, **354**.

[0036] Additionally, the layer **350** may include pads to which the pairs of signal vias **352**, **354** are coupled. For instance, the layer **350** may include a first pad **364**, a second pad **366**, a third pad **368**, and a fourth pad **370**. Respective traces **372** may couple the first signal via **356** and the first pad **364** to one another, the second signal via **358** and the second pad **366** to one another, the third signal via **360** and the third pad **368** to one another, and the fourth signal via **362** and the fourth pad **370** to one another. As an example, the first pad **364** and the second pad **366** may be offset from one another along the first axis **164** and aligned with one another along the second axis **166**, the third pad **368** and the fourth pad **370** may be offset from one another along the first axis **164** and aligned with one another along the second axis **166**, the first pad **364** and the third pad **368** may be aligned with one another along the first axis **164** and offset from one another along the second axis **166**, and the fourth pad **370** and the second pad **366** may be aligned with one another along the first axis **164** and offset from one another along the second axis **166**. In this manner, the pads **364**, **366**, **368**, **370** may form a rectangular (e.g., a square) configuration.

[0037] The pads **364**, **366**, **368**, **370** may help couple the signal vias **356**, **358**, **360**, **362** to a separate component, such as an IC (e.g., using a soldering technique). By way of example, the offset arrangement of the signal vias **356**, **358**, **360**, **362** may not be arranged to align with the pins (e.g., pins having pre-established locations that cannot be adjusted) of the separate component. Rather, the pads **364**, **366**, **368**, **370** may be arranged to align with the pins. As such, the pins may

be configured to mount to the pads **364, 366, 368, 370**, and the traces **372** may then electrically couple the signal vias **356, 358, 360, 362** to the pins by way of the pads **364, 366, 368, 370**. Therefore, the pads **364, 366, 368, 370** may enable the pins and the signal vias **356, 358, 360, 362** to electrically couple to one another despite their misaligned arrangements. That is, the pads **364, 366, 368, 370** are arranged to couple to the pins more readily, such as without having to modify the separate component to move the pins and/or utilize an additional, separate adapter component to electrically couple the signal vias **356, 358, 360, 362** to the pins.

[0038] The traces **372** may extend (e.g., linearly extend) in different directions for the signal vias **352** and for the signal vias **354**. As an example, a first set of traces **372A** may extend from the first pad **364** and from the second pad **366** in a first direction **374** along the first axis **164** toward the first signal via **356** and toward the second signal via **358**, respectively, whereas a second set of traces **372B** may extend from the third pad **368** and from the fourth pad **370** in a second direction **376**, opposite the first direction **374**, toward the third signal via **360** and toward the fourth signal via **362**, respectively. The opposite extension of the first set of traces **372A** and the second set of traces **372B** along the first axis **164** may enable the pads **364, 366, 368, 370** and the signal vias **356, 358, 360, 362** to be arranged relative to one another in an efficient manner, such as without significantly increasing an outer boundary cooperatively occupied by the pads **364, 366, 368, 370** and signal vias **356, 358, 360, 362**. Each of the first set of traces **372A** may form a first angle **378** with the second axis **166**, and each of the second set of traces **372B** may form a second angle **380** with the second axis **166**. In some embodiments, the first angle **378** and the second angle **380** may be substantially different from one another. For example, the first angle **378** may be a value between 40 degrees and 50 degrees (e.g., 45 degrees), whereas the second angle **380** may be less than the first angle **378** and may be a value between 25 degrees and 35 degrees (e.g., 30 degrees). However, in alternative embodiments, the first angle **378** may be less than the second angle **380** or the first angle **378** and the second angle **380** may be substantially equal to one another. Moreover, although each of the traces **372** extends in the same direction along the second axis **166** (e.g., in a downward direction) in the illustrated embodiment, in other embodiments, the traces **372** may extend in different directions along the second axis **166**. For instance, the first set of traces **372A** may extend in a first direction (e.g., a downward direction) along the second axis **166**, whereas the second set of traces **372B** may extend in a second direction (e.g., an upward direction), opposite the first direction, along the second axis **166**. In any case, each of the traces **372** may extend obliquely relative to the first axis **164** and the second axis **166** to arrange the pads **364, 366, 368, 370** and the signal vias **356, 358, 360, 362** in an efficient manner.

[0039] The layer **350** may further include a first ground via **382** positioned adjacent to the first pad **364**, offset from the first pad **364** along the first axis **164**, and aligned with the first pad **364** along the second axis **166**; a second ground via **384** positioned adjacent to the second pad **366**, offset from the second pad **366** along the first axis **164**, and aligned with the second pad **366** along the second axis **166**; a third ground via **386** positioned adjacent to the third pad **368**, offset from the third pad **368** along the first axis **164**, and aligned with the third pad **368** along the second axis **166**; and a fourth ground via **388** positioned adjacent to the fourth pad **370**, offset from the fourth pad **370** along the first axis **164**, and aligned with the fourth pad **370** along the second axis **166**. By way of example, the first pad **382** and the third pad **386** may be aligned with one another along the first axis **164** and the second pad **384** and the fourth pad **388** may be aligned with one another along the first axis **164** to form a rectangular configuration. However, in other embodiments, the first pad **382** and the third pad **386** may be offset from one another along the first axis **164**, and/or the second pad **384** and the fourth pad **388** may be offset from one another along the first axis **164**. In further embodiments, the first ground via **382** may be aligned with the first signal via **356**, instead of the first pad **364**, along the second axis **166**, the second ground via **384** may be aligned with the second signal via **358**, instead of the second pad **366**, along the second axis **166**, the third ground via **386** may be aligned with the third signal via **360**, instead of the third pad **368**, along the second axis

166, and/or the fourth ground via **388** may be aligned with the fourth via **362**, instead of the fourth pad **370**, along the second axis **166**. In any of these embodiments, the first ground via **382** may block electrical fields emitted by the first signal via **356** and/or by the first pad **364**, the second ground via **384** may block electrical fields emitted by the second signal via **358** and/or by the second pad **366**, the third ground via **386** may block electrical fields emitted by the third signal via **360** and/or by the third pad **368**, and/or the fourth ground via **388** may block electrical fields emitted by the fourth signal via **362** and/or by the third pad **370**, thereby isolating such electrical fields (e.g., from other signal vias positioned at the layer **350**).

[0040] FIG. 7 is a flowchart of a method **400** of manufacturing a PCB (e.g., the PCB **100**), such as any of the layers **102**, **150**, **350** discussed herein. In certain embodiments, the method **400** may be performed automatically, such as by a processor executing instructions stored on a memory. In additional or alternative embodiments, the method **400** may be performed manually, such as by an operator. It should be noted that the method **400** may be performed differently than depicted. For example, additional operations may be performed, any of the depicted operations may not be performed, and/or the depicted operations may be performed in a different order.

[0041] At step **402**, a first pair of signal vias may be formed through a layer of a PCB. The first pair of signal vias may include a first signal via and a second signal via offset from one another along a first axis and aligned with one another along a second axis, perpendicular to the first axis. The first signal via and the second signal via may be configured to transmit respective signals. For example, one of the first signal via or the second signal via may be configured to transmit a positive signal, and the other of the first signal via or the second signal via may be configured to transmit a negative signal.

[0042] At step **404**, a second pair of signal vias may be formed through the layer of the PCB. The second pair of signal vias may include a third signal via and a fourth signal via offset from one another along the first axis and aligned with one another along the second axis. Additionally, the second pair of signal vias may be offset from the first pair of signal vias along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from (e.g., greater than) a second distance between the third signal via and the second signal via. For example, an offset distance between the first pair of signal vias and the second pair of signal vias along the first axis may be a value between 0.66 mm and 0.75 mm (e.g., between 0.026 inches and 0.03 inches). Such an offset distance may provide a low overall amount of noise coupling, such as an arrangement in which a first sum of an amount of noise coupling caused by the first signal via with respect to the third signal via and an amount of noise coupling caused by the second signal via with respect to the third signal via is substantially equal to a second sum of an amount of noise coupling caused by the first signal via with respect to the fourth signal via and an amount of noise coupling caused by the second signal via with respect to the fourth signal via. Thus, crosstalk between the pairs of signal vias may be limited.

[0043] In certain embodiments, the offset distance to be established may be determined through experimentation, such as in a calibration phase. For instance, in the calibration phase, the positioning of the signal vias may be adjusted by different offset distances, and corresponding amounts of crosstalk (e.g., noise coupling) may be measured for each offset distance. Thus, a particular offset distance providing the lowest measured amount of crosstalk may be determined during the calibration phase. By way of example, the offset distance that provides the lowest measured amount of crosstalk may vary for different implementations having varying operating parameters/conditions. As such, by measuring the amount of crosstalk for different offset distances, a more suitable offset distance providing the lowest measured amount for a particular implementation may be more accurately determined. Different types of crosstalk, such as near end crosstalk in which noise is produced by a generated signal at where signal transmission originates and far end crosstalk in which noise is produced by a transmitted signal at where the transmitted signal is received, may also be measured. In some cases, for a single implementation, a different

offset distance may produce the lowest amount of different types of crosstalk (e.g., a first offset distance provides the lowest amount of near end crosstalk, whereas a second offset distance, different from the first offset difference, provides the lowest amount of far end crosstalk). In such cases, an intermediate value, such as a mathematical average, between the different offset distances may be utilized.

[0044] At step **406**, ground vias surrounding the pairs of signal vias may be formed through the layer of the PCB. By way of example, a first ground via may be positioned adjacent to the first signal via, offset from the first signal via along the first axis, and aligned with the first signal via along the second axis; a second ground via may be positioned adjacent to the second signal via, offset from the second signal via along the first axis, and aligned with the second signal via along the second axis; a third ground via may be positioned adjacent to the third signal via, offset from the third signal via along the first axis, and aligned with the third signal via along the second axis; and a fourth ground via may be positioned adjacent to the fourth signal via, offset from the fourth signal via along the first axis, and aligned with the fourth signal via along the second axis. For instance, the first ground via and the third ground via may be offset from one another along the first axis and along the second axis, and the second ground via and the fourth ground via may be offset from one another along the first axis and along the second axis. The ground vias may block electrical fields emitted by the pairs of signal vias, thereby reducing crosstalk between the pairs of signal vias and other pairs of signal vias at the layer of the PCB.

[0045] In some embodiments, the signal vias are configured to electrically couple to pins of a separate component. As an example, the pins of the separate component may be arranged in a similar manner as that of the signal vias. Thus, the signal vias may align with and directly engage with (e.g., using a soldering technique) the pins to electrically couple to the pins. As another example, the pins of the separate component may be arranged in a different manner than that of the signal vias. For this reason, the signal vias may not be able to align with the pins. Instead, pads may be formed at the layer of the PCB and arranged to align with the pins, and respective traces may be formed at the layer of the PCB to electrically couple the pads and signal vias to one another. Therefore, the pads may be configured to couple to the pins (e.g., using a soldering technique), and the pins and signal vias may be electrically coupled to one another by way of the traces and the pads. In either embodiment, the PCB and the separate component may communicate with one another using signals transmitted through the signal vias.

[0046] In some aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; a first pair of signal vias formed through the layer, wherein the first pair of signal vias is configured to transmit respective signals, the first pair of signal vias includes a first signal via and a second signal via offset from one another along a first axis and aligned with one another along a second axis, perpendicular to the first axis; and a second pair of signal vias formed through the layer, wherein the second pair of signal vias is configured to transmit respective signals, the second pair of signal vias includes a third signal via and a fourth signal via offset from one another along the first axis and aligned with one another along the second axis, and the first pair of signal vias and the second pair of signal vias are offset from one another along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from a second distance between the third signal via and the second signal via.

[0047] In some aspects, the techniques described herein relate to an apparatus, wherein the first distance between the third signal via and the first signal via is greater than the second distance between the third signal via and the second signal via.

[0048] In some aspects, the techniques described herein relate to an apparatus, wherein the first pair of signal vias and the second pair of signal vias are offset from one another along the first axis and along the second axis such that a first sum of a first amount of noise coupling caused by the first signal via with respect to the third signal via plus a second amount of noise coupling caused by the second signal via with respect to the third signal via is substantially equal to a second sum of a

third amount of noise coupling caused by the first signal via with respect to the fourth signal via plus a fourth amount of noise coupling caused by the second signal via with respect to the fourth signal via.

[0049] In some aspects, the techniques described herein relate to an apparatus, wherein the first pair of signal vias and the second pair of signal vias are configured to mount to pins of an integrated circuit.

[0050] In some aspects, the techniques described herein relate to an apparatus, further including: a plurality of pads formed on the layer of the printed circuit board and configured to mount to pins of an integrated circuit; and a plurality of traces that connects the first signal via, the second signal via, the third signal via, and the fourth signal via to a respective one of the plurality of pads.

[0051] In some aspects, the techniques described herein relate to an apparatus, wherein the plurality of traces includes: a first set of traces that connects the first signal via and the second signal via to respective first pads of the plurality of pads, wherein each trace of the first set of traces extends in a first direction along the first axis; and a second set of traces that connects the third signal via and the fourth signal via to respective second pads of the plurality of pads, wherein each trace of the second set of traces extends in a second direction, opposite the first direction, along the first axis.

[0052] In some aspects, the techniques described herein relate to an apparatus, wherein an offset distance between the first signal via and the third signal via along the first axis is between 0.66 millimeters and 0.75 millimeters.

[0053] In some aspects, the techniques described herein relate to an apparatus, further including a first ground via and a second ground via offset from the first pair of signal vias along the first axis and aligned with the first pair of signal vias along the second axis, wherein a third distance between the first signal via and the first ground via, a fourth distance between the first signal via and the second signal via, and a fifth distance between the second signal via and the second ground via are substantially equal to one another.

[0054] In some aspects, the techniques described herein relate to an apparatus including: a layer of a printed circuit board; and a plurality of signal vias formed through the layer and configured to transmit respective signals, wherein the plurality of signal vias includes: a first signal via; a second signal via immediately adjacent to the first signal via along a first axis; a third signal via immediately adjacent to the first signal via along a second axis, perpendicular to the first axis; and a fourth signal via immediately adjacent to the third signal via along the first axis, wherein the first signal via, the second signal via, the third signal via, and the fourth signal via are arranged in a parallelogram configuration, and a first distance between the first signal via and the third signal via is different from a second distance between the second signal via and the third signal via.

[0055] In some aspects, the techniques described herein relate to an apparatus, wherein the first distance between the first signal via and the third signal via is greater than the second distance between the second signal via and the third signal via.

[0056] In some aspects, the techniques described herein relate to an apparatus, wherein the second distance between the second signal via and the third signal via is less than a third distance between the first signal via and the second signal via.

[0057] In some aspects, the techniques described herein relate to an apparatus, wherein the third distance between the first signal via and the second signal via is substantially equal to a fourth distance between the third signal via and the fourth signal via.

[0058] In some aspects, the techniques described herein relate to an apparatus, wherein the third distance between the first signal via and the second signal via is substantially equal to a fourth distance between the first signal via and the third signal via along the second axis.

[0059] In some aspects, the techniques described herein relate to an apparatus, wherein the third signal via is positioned more adjacent to the second signal via than to the first signal via along the first axis.

[0060] In some aspects, the techniques described herein relate to an apparatus including: a layer of

a printed circuit board; and a plurality of signal vias formed through the layer and configured to transmit respective signals, wherein the plurality of signal vias includes: a first signal via; a second signal via offset from the first signal via along a first axis and aligned with the first signal via along a second axis, perpendicular to the first axis; a third signal via offset from the first signal via and the second signal via along the first axis and along the second axis; and a fourth signal via offset from the third signal via along the first axis and aligned with the third signal via along the second axis, wherein a first distance between the first signal via and the third signal via, a second distance between the first signal via and the second signal via, and a third distance between the second signal via and the third signal via are different from one another, and the second distance between the first signal via and the second signal via is substantially equal to a fourth distance between the third signal via and the fourth signal via.

[0061] In some aspects, the techniques described herein relate to an apparatus, wherein each of the first distance and the second distance is greater than the third distance.

[0062] In some aspects, the techniques described herein relate to an apparatus, further including: a plurality of pads formed on the layer of the printed circuit board and configured to mount to pins of an integrated circuit; a first trace that connects the first signal via to a first pad of the plurality of pads; a second trace that connects the third signal via to a second pad of the plurality of pads, wherein the first trace forms a first angle with the second axis, and the second trace forms a second angle, different from the first angle, with the second axis.

[0063] In some aspects, the techniques described herein relate to an apparatus, wherein the second angle is less than the first angle.

[0064] In some aspects, the techniques described herein relate to an apparatus, further including: a fifth signal via offset from the first signal via and the third signal via along the second axis and aligned with the first signal via along the first axis; and a sixth signal via offset from the second signal via and the fourth signal via along the second axis and aligned with the second signal via along the first axis.

[0065] In some aspects, the techniques described herein relate to an apparatus, wherein the first signal via and the second signal via are a first differential pair, and the third signal via and the fourth signal via are a second differential pair.

[0066] The above description is intended by way of example only. Although the techniques are illustrated and described herein as embodied in one or more specific examples, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made within the scope and range of equivalents of the claims.

[0067] As used herein, unless expressly stated to the contrary, use of the phrase 'at least one of', 'one or more of', 'and/or', variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions 'at least one of X, Y and Z', 'at least one of X, Y or Z', 'one or more of X, Y and Z', 'one or more of X, Y or Z' and 'X, Y and/or Z' can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.

[0068] Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in 'one embodiment', 'example embodiment', 'an embodiment', 'another embodiment', 'certain embodiments', 'some embodiments', 'various embodiments', 'other embodiments', 'alternative embodiment', and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments.

[0069] Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound

embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

[0070] Additionally, unless expressly stated to the contrary, the terms ‘first’, ‘second’, ‘third’, etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, ‘first X’ and ‘second X’ are intended to designate two ‘X’ elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, ‘at least one of’ and ‘one or more of’ can be represented using the ‘(s)’ nomenclature (e.g., one or more element(s)).

[0071] As used herein, the terms “approximately,” “generally,” “substantially,” and so forth, are intended to convey that the property value being described may be within a relatively small range of the property value, as those of ordinary skill would understand. For example, when a property value is described as being “approximately” equal to (or, for example, “substantially similar” to) a given value, this is intended to convey that the property value may be within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, of the given value. Similarly, when a given feature is described as being “substantially parallel” to another feature, “generally perpendicular” to another feature, and so forth, this is intended to convey that the given feature is within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, to having the described nature, such as being parallel to another feature, being perpendicular to another feature, and so forth. Mathematical terms, such as “parallel” and “perpendicular,” should not be rigidly interpreted in a strict mathematical sense, but should instead be interpreted as one of ordinary skill in the art would interpret such terms. For example, one of ordinary skill in the art would understand that two lines that are substantially parallel to each other are parallel to a substantial degree, but may have minor deviation from exactly parallel.

[0072] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible, or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . or “step for [perform]ing [a function] . . .”, it is intended that such elements are to be interpreted under 35 U.S.C. 112 (f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112 (f).

[0073] One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

Claims

1. An apparatus comprising: a layer of a printed circuit board; a first pair of signal vias formed through the layer, wherein the first pair of signal vias is configured to transmit respective signals, the first pair of signal vias comprises a first signal via and a second signal via offset from one another along a first axis and aligned with one another along a second axis, perpendicular to the first axis; and a second pair of signal vias formed through the layer, wherein the second pair of signal vias is configured to transmit respective signals, the second pair of signal vias comprises a third signal via and a fourth signal via offset from one another along the first axis and aligned with one another along the second axis, and the first pair of signal vias and the second pair of signal vias

are offset from one another along the first axis and along the second axis such that a first distance between the third signal via and the first signal via is different from a second distance between the third signal via and the second signal via.

2. The apparatus of claim 1, wherein the first distance between the third signal via and the first signal via is greater than the second distance between the third signal via and the second signal via.
3. The apparatus of claim 1, wherein the first pair of signal vias and the second pair of signal vias are offset from one another along the first axis and along the second axis such that a first sum of a first amount of noise coupling caused by the first signal via with respect to the third signal via plus a second amount of noise coupling caused by the second signal via with respect to the third signal via is substantially equal to a second sum of a third amount of noise coupling caused by the first signal via with respect to the fourth signal via plus a fourth amount of noise coupling caused by the second signal via with respect to the fourth signal via.
4. The apparatus of claim 1, wherein the first pair of signal vias and the second pair of signal vias are configured to mount to pins of an integrated circuit.
5. The apparatus of claim 1, further comprising: a plurality of pads formed on the layer of the printed circuit board and configured to mount to pins of an integrated circuit; and a plurality of traces that connects the first signal via, the second signal via, the third signal via, and the fourth signal via to a respective one of the plurality of pads.
6. The apparatus of claim 5, wherein the plurality of traces comprises: a first set of traces that connects the first signal via and the second signal via to respective first pads of the plurality of pads, wherein each trace of the first set of traces extends in a first direction along the first axis; and a second set of traces that connects the third signal via and the fourth signal via to respective second pads of the plurality of pads, wherein each trace of the second set of traces extends in a second direction, opposite the first direction, along the first axis.
7. The apparatus of claim 1, wherein an offset distance between the first signal via and the third signal via along the first axis is between 0.66 millimeters and 0.75 millimeters.
8. The apparatus of claim 1, further comprising a first ground via and a second ground via offset from the first pair of signal vias along the first axis and aligned with the first pair of signal vias along the second axis, wherein a third distance between the first signal via and the first ground via, a fourth distance between the first signal via and the second signal via, and a fifth distance between the second signal via and the second ground via are substantially equal to one another.
9. An apparatus comprising: a layer of a printed circuit board; and a plurality of signal vias formed through the layer and configured to transmit respective signals, wherein the plurality of signal vias comprises: a first signal via; a second signal via immediately adjacent to the first signal via along a first axis; a third signal via immediately adjacent to the first signal via along a second axis, perpendicular to the first axis; and a fourth signal via immediately adjacent to the third signal via along the first axis, wherein the first signal via, the second signal via, the third signal via, and the fourth signal via are arranged in a parallelogram configuration, and a first distance between the first signal via and the third signal via is different from a second distance between the second signal via and the third signal via.
10. The apparatus of claim 9, wherein the first distance between the first signal via and the third signal via is greater than the second distance between the second signal via and the third signal via.
11. The apparatus of claim 9, wherein the second distance between the second signal via and the third signal via is less than a third distance between the first signal via and the second signal via.
12. The apparatus of claim 11, wherein the third distance between the first signal via and the second signal via is substantially equal to a fourth distance between the third signal via and the fourth signal via.
13. The apparatus of claim 11, wherein the third distance between the first signal via and the second signal via is substantially equal to a fourth distance between the first signal via and the third signal via along the second axis.

- 14.** The apparatus of claim 9, wherein the third signal via is positioned more adjacent to the second signal via than to the first signal via along the first axis.
- 15.** An apparatus comprising: a layer of a printed circuit board; and a plurality of signal vias formed through the layer and configured to transmit respective signals, wherein the plurality of signal vias comprises: a first signal via; a second signal via offset from the first signal via along a first axis and aligned with the first signal via along a second axis, perpendicular to the first axis; a third signal via offset from the first signal via and the second signal via along the first axis and along the second axis; and a fourth signal via offset from the third signal via along the first axis and aligned with the third signal via along the second axis, wherein a first distance between the first signal via and the third signal via, a second distance between the first signal via and the second signal via, and a third distance between the second signal via and the third signal via are different from one another, and the second distance between the first signal via and the second signal via is substantially equal to a fourth distance between the third signal via and the fourth signal via.
- 16.** The apparatus of claim 15, wherein each of the first distance and the second distance is greater than the third distance.
- 17.** The apparatus of claim 15, further comprising: a plurality of pads formed on the layer of the printed circuit board and configured to mount to pins of an integrated circuit; a first trace that connects the first signal via to a first pad of the plurality of pads; and a second trace that connects the third signal via to a second pad of the plurality of pads, wherein the first trace forms a first angle with the second axis, and the second trace forms a second angle, different from the first angle, with the second axis.
- 18.** The apparatus of claim 17, wherein the second angle is less than the first angle.
- 19.** The apparatus of claim 15, further comprising: a fifth signal via offset from the first signal via and the third signal via along the second axis and aligned with the first signal via along the first axis; and a sixth signal via offset from the second signal via and the fourth signal via along the second axis and aligned with the second signal via along the first axis.
- 20.** The apparatus of claim 15, wherein the first signal via and the second signal via are a first differential pair, and the third signal via and the fourth signal via are a second differential pair.
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